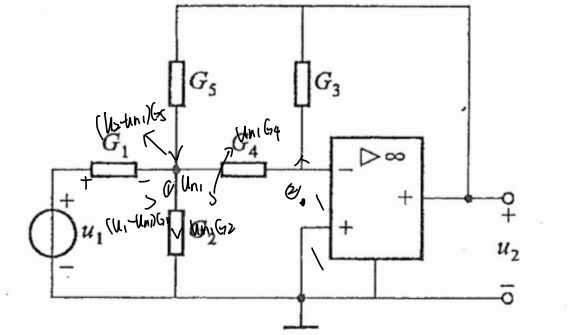
5-3 求题 5-3 图所示电路的输出电压与输入电压之比$\frac{u_2}{u_1}$。



题 5-3 图

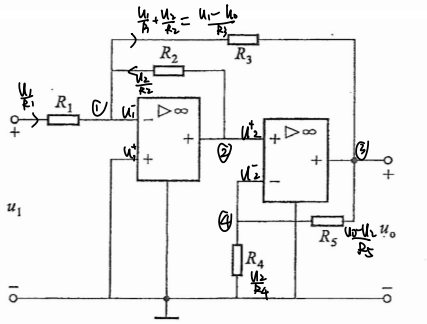
$u^+=u^-=0$ ①,②节点电压法

节点①KCL: $(u_1-u_{n1})G_1+(u_2-u_{n1})G_5=u_{n1}G_2+u_{n1}G_4$

节点②KCL: $u_{n1}G_4=-u_2G_3 \Rightarrow u_{n1}=-\frac{G_3}{G_4}u_2$

$\Rightarrow \frac{u_2}{u_1}=-\frac{G_1G_4}{G_1G_3+G_3G_5+G_2G_3+G_3G_4+G_4G_5}$

5-4 求题 5-4 图所示电路的电压比值$\frac{u_o}{u_1}$。



题 5-4 图

$u_1^-=u_1^+=0 \Rightarrow u_{n1}=0$

①:KCL $\frac{u_1}{R_1}+\frac{u_2}{R_2}=\frac{u_1-u_o}{R_3}$

④:KCL: $\frac{u_o-u_2}{R_5}=\frac{u_2}{R_4}$ $\Rightarrow \frac{u_o}{u_1}=\frac{R_2R_3(R_4+R_5)}{R_1(R_3R_4+R_3R_4+R_2R_5)}$

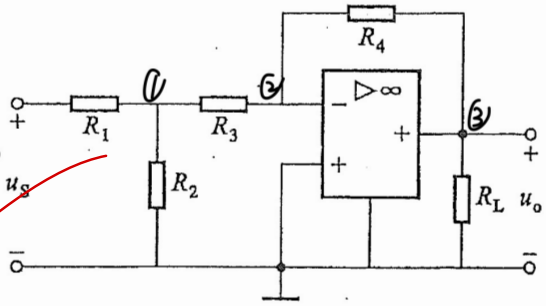
5-5 求题 5-5 图所示电路的电压比$\frac{u_o}{u_S}$。

$u^-=u^+=0 \Rightarrow u_{n2}=0$

$(\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3})u_{n1}-\frac{1}{R_3}u_{n2}=\frac{u_S}{R_1}$

$-\frac{1}{R_3}u_{n1}+(\frac{1}{R_3}+\frac{1}{R_4})u_{n2}-\frac{1}{R_4}u_o=0$

$\Rightarrow \frac{u_o}{u_S}=\frac{-R_2R_4}{R_2R_3+R_1R_3+R_1R_2}$



题 5-5 图

5-7 求题 5-7 图所示电路的 u_o 与 u_{S1}、u_{S2} 之间的关系。

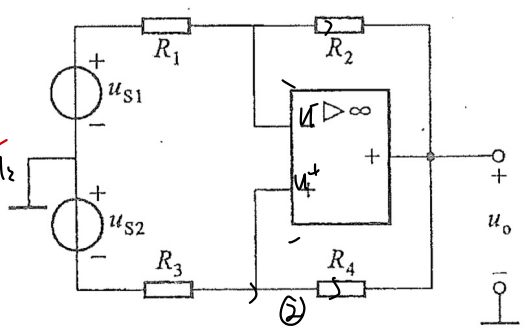
①:KCL: $\frac{u_{S1}-u^-}{R_1}=\frac{u^--u_o}{R_2}$

②KCL: $\frac{-u_{S2}-u^+}{R_3}=\frac{u^+-u_o}{R_4}$

$u^-=u^+$

$\Rightarrow \frac{R_2R_3-R_1R_4}{R_3R_4(R_1+R_2)}u_o=\frac{R_2(R_3+R_4)}{R_3R_4(R_1+R_2)}u_1+\frac{1}{R_3}u_2$

$\Rightarrow u_o=\frac{R_2(R_3+R_4)u_{S1}+R_4(R_1+R_2)u_{S2}}{R_2R_3-R_1R_4}$



题 5-7 图